FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 1**

General Instructions:
1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt any 11 questions. For each, choose either the main question OR its alternative.
4. In Section C, attempt all 4 long questions. For each, choose either the main question OR its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Who first introduced binomial nomenclature and is called the "Father of Taxonomy"?	Aristotle	Carl Woese	Carolus Linnaeus	Robert Whittaker	○○○○
2	The three-domain system of classification was proposed by:	Carolus Linnaeus	Carl Woese	Robert Whittaker	Robert Hooke	○○○○
3	Which organelle is called the "powerhouse of the cell" because it produces ATP?	Ribosome	Golgi apparatus	Lysosome	Mitochondria	○○○○
4	The longest phase of mitosis, during which chromosomes first become visible, is:	Metaphase	Anaphase	Prophase	Telophase	○○○○
5	Which lists the levels of body organization correctly from smallest to largest?	Organism, organ system, organ, tissue, cell	Tissue, cell, organ, organ system, organism	Cell, tissue, organ, organ system, organism	Organ, organ system, organism, tissue, cell	○○○○
6	The tendency to maintain a stable, relatively constant internal environment is called:	Osmoregulation	Thermoregulation	Homeostasis	Metabolism	○○○○
7	Which of the following is a MAJOR bioelement constituting 99% of protoplasm?	Sodium	Phosphorus	Iron	Magnesium	○○○○
8	Peptide bonds are formed between two:	Monosaccharides	Amino acids	Nucleotides	Fatty acids	○○○○
9	Enzymes are biologically active globular proteins that:	Store hereditary information	Tremendously speed up biochemical reactions	Provide long-term energy to cells	Transport oxygen in blood	○○○○
10	In competitive inhibition, the inhibitor occupies the:	Allosteric site	Active site	Prosthetic group	Coenzyme	○○○○
11	ATP molecule is made up of Adenine, Ribose sugar, and:	Two phosphate groups	One phosphate group	Three phosphate groups	Four phosphate groups	○○○○
12	The vector responsible for transmitting Malaria is the:	Housefly	Female Aedes mosquito	Female Anopheles mosquito	Culex mosquito	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define Biology. State THREE major fields of Biology with one example organism each.	3	**OR**	What is the biological method? List its six steps in correct sequence.	3
2(ii)	Differentiate between hypothesis, theory, and law with one example each.	3	**OR**	Describe the role of Ronald Ross in proving that Malaria is transmitted by the Anopheles mosquito.	3
2(iii)	What is binomial nomenclature? Give THREE rules for writing a scientific name with an example.	3	**OR**	Distinguish between the three domains of Carl Woese, naming the basis of his classification.	3

2(iv)	State THREE functions of the cell membrane and give its chemical composition (fluid mosaic model).	3	**OR**	Compare smooth endoplasmic reticulum (SER) with rough endoplasmic reticulum (RER) in structure and function.	3
2(v)	Explain the significance of mitosis under three headings: genetic stability, growth, and wound healing.	3	**OR**	Differentiate between cytokinesis in animal cells and plant cells during cell division.	3
2(vi)	What is crossing over? State where it occurs in meiosis and explain its importance for genetic variation.	3	**OR**	Distinguish between benign and malignant tumours. Define metastasis.	3
2(vii)	How do different tissues form the stomach? Name the four tissue types in its wall and their roles.	3	**OR**	Explain how the respiratory system and cardiovascular system cooperate to exchange gases in the body.	3
2(viii)	Describe the structure and function of the three types of plant tissue: dermal, ground, and vascular.	3	**OR**	Explain the structure of a leaf in cross-section, naming palisade mesophyll, spongy mesophyll, guard cells, xylem and phloem.	3
2(ix)	Compare energy yield per gram of lipids, carbohydrates, and proteins. Name THREE functions of lipids in living organisms.	3	**OR**	Classify carbohydrates into three groups with one example and one function each.	3
2(x)	State THREE main points of the Watson and Crick model of DNA structure.	3	**OR**	Name the three types of RNA and state the specific role of each in protein synthesis.	3
2(xi)	Define enzyme inhibition. Distinguish between competitive and non-competitive inhibition.	3	**OR**	Name FOUR factors affecting enzyme activity and explain how temperature affects enzyme functioning.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Give a detailed account of the levels of biological organization from atomic level to organism level, with explanation and one example at each level.	5	**OR**	Describe the advantages of classification. Write the complete taxonomic hierarchy from domain to species, using the human being as an example at each level.	5
3(ii)	Write a detailed note on the structure and functions of: (a) Mitochondria (b) Chloroplast (c) Golgi apparatus.	5	**OR**	Describe the complete process of mitosis, covering all four stages of karyokinesis and cytokinesis, noting differences between plant and animal cells.	5
3(iii)	Describe the structure, functions, and sources of proteins. Include amino acids, peptide bonds, and the four structural levels of proteins.	5	**OR**	Explain the central dogma of molecular biology. Describe transcription and translation in detail, naming all molecules involved.	5
3(iv)	Explain the mechanism of enzyme action (ES/EP complex), the characteristics of enzymes, and how enzyme activity is regulated by inhibitors and activators.	5	**OR**	Define ATP and explain its role as energy currency. Describe the ATP-ADP cycle, the three components of the ATP molecule, and the role of high-energy bonds.	5

Invigilator's Signature: ____________________ Candidate's Name: ____________________ Roll No: __________

Chapters: Ch 1 (Science of Biology) | Ch 2 (Biodiversity) | Ch 3 (The Cell) | Ch 4 (Cell Cycle) | Ch 5 (Tissues, Organs & Organ Systems) | Ch 6 (Molecular Biology) | Ch 7 (Metabolism Sec 7.1–7.6)

— End of Question Paper —